

INTERVIEW PACK

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RECOMMENDATION HIRE	DURATION 60 min	TECHNICAL 7 questions	BEHAVIORAL 5 questions	RISKS FLAGGED 5
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01 EXECUTIVE SUMMARY

Shrijal Goswami, 1.5 years of hands-on ML/AI engineering experience with end-to-end NLP and predictive modeling projects, currently a B.Tech student

WHY SHORTLISTEDStrong practical experience building, evaluating, and deploying ML models (scikit-learn, XGBoost, PyTorch) and full-stack AI products; matches core intern responsibilities despite minor tooling gaps

- DIFFERENTIATORS
- Built a hybrid BM25 + dense retrieval pipeline achieving MRR0.84
 - Deployed stacking-ensemble heart-disease model via FastAPI/Docker with 0.94 ROC-AUC
 - Designed multi-LLM booklet generation engine reducing token cost≈ 10×

02 INTERVIEW PLAN

INTERVIEWERSenior ML Engineer with MLOps experience, familiar with retrieval systems and containerized deployments

#	STAGE	TIME	FOCUS
1	Intro & Fit	10 min	Confirm motivation, availability, and cultural alignment
2	Technical Deep-Dive	35 min	Assess ML fundamentals, system design, and production experience
3	Behavioral & Leadership	15 min	Explore ownership, decision-making, and communication

- PRIORITY FOCUS
- Hybrid retrieval architecture
 - Model evaluation & tuning
 - Production deployment (FastAPI/Docker)
 - Experiment-tracking design

03 TECHNICAL QUESTIONS

T1 Can you describe the feature-engineering steps you applied in the Heart Disease Prediction System?		EASY	FEATURE ENGINEERING
WHY ASK	Validate basic data-preparation knowledge required for the internship		
STRONG ANSWER	Handling missing values, encoding categorical variables, scaling numeric features, possibly creating interaction terms or domain-specific ratios		
RED FLAGS	Vague answer, no mention of specific techniques		
EVALUATE ON	- Clarity of steps - Justification of choices - Link to performance gain		
FOLLOW-UPS	- Why did you choose those transformations? - How did you verify they improved model performance?		
T2 Walk me through the architecture of the hybrid retrieval pipeline you built for HireLens, specifically how you combined BM25 and dense embeddings.		MEDIUM	RETRIEVAL SYSTEM DESIGN
WHY ASK	Directly ties to candidate's most complex project and tests system-level thinking		
STRONG ANSWER	Data ingestion → text split → BM25 index creation → sentence-transformer embeddings → dense index (e.g., ChromaDB) → at query time retrieve BM25 scores and dense scores, combine with weighted sum (α=0.65), rerank, return top-k		
RED FLAGS	Missing weighting detail, no mention of indexing or latency considerations		
EVALUATE ON	- Depth of architectural detail - Understanding of sparse vs dense trade-offs - Awareness of latency and scalability		
FOLLOW-UPS	- What trade-offs did you consider when setting α? - How did you handle updates to the corpus?		

T3 How did you perform hyper-parameter tuning for the XGBoost model in the stacking ensemble for heart-disease prediction? Include your validation strategy.

HARD

MODEL TUNING

WHY ASK	Assess rigorous experimentation and awareness of over-fitting, a key intern responsibility
STRONG ANSWER	Used randomized search over max_depth, learning_rate, n_estimators; 5-fold stratified cross-validation; evaluated using ROC-AUC; selected parameters that maximized validation AUC while monitoring training vs validation gap
RED FLAGS	No cross-validation, only single train-test split
EVALUATE ON	- Methodology correctness - Metric selection - Evidence of preventing over-fit
FOLLOW-UPS	- Did you use early stopping? - How did you handle class imbalance?

T4 Design an experiment-tracking solution for your multi-LLM Booklet Generation Engine that ensures reproducibility and can scale to many concurrent users.

EXPERT

MLOPS / EXPERIMENT TRACKING

WHY ASK	Candidate lacks explicit experience with tracking tools; tests ability to design appropriate solution
STRONG ANSWER	Use MLflow or Weights & Biases to log parameters, prompt templates, model versions, random seeds; store artifacts in cloud bucket; integrate with CI/CD pipeline; tag runs with user ID; provide UI for comparison; ensure Docker image versioning for reproducibility
RED FLAGS	No mention of versioning, metadata, or scalability
EVALUATE ON	- Comprehensiveness of logging - Scalability considerations - Reproducibility mechanisms
FOLLOW-UPS	- How would you handle secret management for API keys? - What would you store to debug a failed generation?

T5 Explain how you containerized and deployed the Heart Disease Prediction model using FastAPI and Docker, and how you addressed concurrent request handling.

MEDIUM

MODEL DEPLOYMENT

WHY ASK	Matches internship's deployment expectations
STRONG ANSWER	Created a FastAPI app with a /predict endpoint, loaded serialized model at startup, used Uvicorn workers (e.g., gunicorn with multiple workers) for concurrency, built Dockerfile copying code and model, exposed port 80, tested with Docker Compose; added health-check endpoint
RED FLAGS	Single-threaded approach, no mention of workers
EVALUATE ON	- Correctness of deployment steps - Concurrency handling - Awareness of monitoring
FOLLOW-UPS	- How would you monitor latency in production? - What would you do if model file size exceeds container limits?

T6 Which evaluation metrics would you choose for a binary classification model in a medical diagnosis context and why?

EASY

MODEL EVALUATION

WHY ASK	Ensures candidate knows appropriate metrics beyond accuracy
STRONG ANSWER	ROC-AUC for overall discrimination, precision-recall curve due to class imbalance, F1-score for balance, confusion matrix to inspect false negatives (critical in medical), possibly calibration metrics
RED FLAGS	Only mentions accuracy
EVALUATE ON	- Metric relevance - Understanding of domain impact
FOLLOW-UPS	How would you prioritize false negatives vs false positives?

T7 Given a highly imbalanced dataset, what techniques would you apply when training a scikit-learn model to improve performance?

HARD

IMBALANCED DATA HANDLING

WHY ASK	Tests depth of practical ML knowledge
STRONG ANSWER	Resampling (SMOTE, undersampling), class_weight='balanced', threshold tuning, using metrics like ROC-AUC, possibly ensemble methods, stratified cross-validation
RED FLAGS	No concrete technique
EVALUATE ON	- Range of techniques mentioned - Justification for chosen method
FOLLOW-UPS	Which method would you pick for XGBoost and why?

04 BEHAVIORAL QUESTIONS

B1 In the AKademyOS project you acted as the architect. Describe a situation where you had to lead a technical decision and how you handled dissent from teammates.		LEADERSHIP
WHY ASK	Validate real leadership experience and conflict resolution	
STRONG ANSWER	Explain the decision (e.g., choosing Nextjs + FastAPI), present opposing view, describe data-driven discussion, consensus building, and final outcome	
WARNING SIGNS	No team involvement, solo decision	

B2 Tell me about a disagreement you encountered while building the Resume Intelligence Platform at Pinnacle Labs and how you resolved it.		CONFLICT
WHY ASK	Assess ability to manage technical disagreements	
STRONG ANSWER	Identify the disagreement (e.g., storage choice), steps taken (meeting, prototype comparison), resolution and lessons learned	
WARNING SIGNS	Blames others, no resolution	

B3 Give an example where you identified a production issue in the ASK AK doubt-solving engine and took ownership to fix it.		OWNERSHIP
WHY ASK	Check proactive problem-solving	
STRONG ANSWER	Describe the issue (e.g., rate-limit breach), diagnostics, patch (caching, back-off), monitoring added	
WARNING SIGNS	No concrete example	

B4 Describe a situation where you had to choose between building a custom retrieval component versus using an existing library for the hybrid search pipeline. What factors influenced your decision?		DECISION-MAKING
WHY ASK	Evaluate analytical decision process	
STRONG ANSWER	Factors: performance, flexibility, maintenance, time-to-market; outcome and justification	
WARNING SIGNS	Decision based solely on convenience	

B5 How did you explain the technical details of the stacking ensemble model to non-technical stakeholders in the Heart Disease Prediction project?		COMMUNICATION
WHY ASK	Assess ability to translate complex ML concepts	
STRONG ANSWER	Used analogies, visual ROC curves, highlighted business impact, avoided jargon	
WARNING SIGNS	Unable to simplify	

05 SKILL VERIFICATION

SKILL	HOW TO VERIFY	CONFIDENCE
Python	Hands-on coding exercise EXERCISE Write a function that reads a CSV, fills missing numeric values with the median, encodes categorical columns with one-hot, and returns a pandas DataFrame DISCUSS Typical data-wrangling workflow in ML projects	Medium
Scikit-learn / XGBoost	Discussion & whiteboard DISCUSS Design and evaluate a gradient-boosted model for a binary classification task, covering train-test split, hyper-parameter tuning, and metric selection	High
FastAPI & Docker	Live coding EXERCISE Create a minimal FastAPI app that loads a pickled model and returns predictions; containerize it with a Dockerfile and run locally DISCUSS Concurrency handling and health-checks	High
Hybrid Retrieval (BM25 + Dense)	System design discussion DISCUSS Explain indexing, query processing, weighting, and latency considerations	Medium
Git/GitHub	Discussion DISCUSS Branching strategy for collaborative ML projects, handling merge conflicts, and CI integration	High

06 RISK ASSESSMENT

Skill Gap

SIGNAL	Candidate has not provided concrete evidence of using NumPy/Pandas for data preprocessing
INVESTIGATE	Ask detailed data-wrangling questions and request code samples

Tooling Gap

SIGNAL	No explicit experience with systematic experiment-tracking tools (MLflow, W&B)
INVESTIGATE	Probe design of experiment-tracking solution and past practices

Cloud Deployment

SIGNAL	Deployment experience limited to Docker containers; minimal exposure to major cloud platforms (AWS/GCP/Azure)
INVESTIGATE	Discuss any work with Vertex AI or cloud services and assess depth

Scale Experience

SIGNAL	Production traffic limited to ~35 active users; may lack experience handling high-throughput loads
INVESTIGATE	Explore how candidate would scale FastAPI service to thousands of concurrent requests

Academic Commitment

SIGNAL	Currently enrolled in a B.Tech program (2024-2028); internship may compete with coursework
INVESTIGATE	Confirm availability and time-management plan for the internship

07 INTERVIEWER SCORECARD

CATEGORY	WEIGHT	FOCUS	SCORE
Technical	30	Depth of ML algorithms, retrieval architecture, deployment stack	
Problem Solving	20	Hyper-parameter tuning, handling imbalanced data, scaling pipelines	
Communication	15	Clarity in explaining models to non-technical audience, documentation habits	
Leadership	10	Architectural decisions, leading solo projects, influencing teammates	
Learning Ability	10	Adoption of new tools (experiment tracking, cloud services), self-learning from certifications	
Culture Fit	10	Collaboration in remote teams, openness to feedback, alignment with NovaScale AI values	
Ownership	5	Proactive issue detection, end-to-end delivery, post-deployment monitoring	

08 RECOMMENDATION

HIRE

Candidate demonstrates solid hands-on ML experience, end-to-end product delivery, and relevant tech stack (Python, scikit-learn, XGBoost, PyTorch, FastAPI, Docker). Gaps (experiment tracking, cloud deployment) are addressable at the intern level and can be mitigated through mentorship. Overall fit with role responsibilities is strong.

UNCERTAINTY Moderate uncertainty around large-scale production readiness and depth of data-wrangling tool usage